

$\Phi(t, x, y)$, and pressure perturbation $\delta P(t, x, y)$, viz.

$$u^x = U(y) + \partial_y \Phi(t, x, y), \quad (\text{A.2})$$

$$u^y = 0 - \partial_x \Phi(t, x, y), \quad (\text{A.3})$$

$$u^z = 0, \quad (\text{A.4})$$

$$P = -2\nu\rho x + \delta P(t, x, y), \quad (\text{A.5})$$

where $U(y) = 1 - y^2$. The perturbed flow is incompressible, $\vec{\nabla} \cdot \vec{u} = 0$, and we impose no-slip boundary conditions $\Phi(t, \pm 1) = \partial_y \Phi(t, \pm 1) = 0$. In order to correctly account for energy in the system we work to second order in perturbation theory,

$$\delta P = \epsilon P^{(1)}(t, x, y) + \epsilon^2 P^{(2)}(t, x, y) + O(\epsilon)^3 \quad (\text{A.6})$$

$$\Phi = \epsilon \Phi^{(1)}(t, x, y) + \epsilon^2 \Phi^{(2)}(t, x, y) + O(\epsilon)^3, \quad (\text{A.7})$$

where ϵ is a formal parameter counting orders in the expansion. Let us consider each order in turn.

At $O(\epsilon)$ we take a real perturbation formed as follows:

$$\Phi^{(1)}(t, x, y) = \phi(t, y)e^{i\alpha x} + \bar{\phi}(t, y)e^{-i\alpha x} \quad (\text{A.8})$$

with wavenumber $\alpha \neq 0$. $P^{(1)}$ is determined algebraically by the x -component of (A.1) at order ϵ . The y component of (A.1), after inserting $P^{(1)}$, gives the Orr-Sommerfeld equation (2.1), where the Orr-Sommerfeld operator is

$$O_{\text{OS}} = -\Delta_2^{-1} [(i\alpha Re)^{-1} \Delta_2^2 - U(y)\Delta_2 + U''(y)], \quad (\text{A.9})$$

with Reynolds number $Re = \nu^{-1}$, and where $\Delta_2 = \partial_y^2 - \alpha^2$ is the spatial Laplacian for the x, y -plane.

At $O(\epsilon)^2$ we have perturbations sourced by $O(\epsilon)$ terms. These take the form,

$$\Phi^{(2)}(t, x, y) = \delta\phi_0(t, y) + \sum_{\pm} \delta\phi_{\pm\alpha}(t, y)e^{\pm i2\alpha x}. \quad (\text{A.10})$$

The zero-momentum piece $\partial_y \delta\phi_0$ obeys the following diffusion equation,

$$(\partial_t - \nu\partial_y^2) \partial_y \delta\phi_0 = i\alpha\partial_y (\phi\partial_y \bar{\phi} - \bar{\phi}\partial_y \phi), \quad (\text{A.11})$$

with viscosity ν serving as the diffusivity, and a current source term coming from the $O(\epsilon)$ perturbations. $P^{(2)}$, $\delta\phi_{\pm\alpha}(t, y)$ are determined by other differential equations but we will not need them here.

To $O(\epsilon)^2$ the energy of the perturbed flow evaluates to

$$E = \int_{-1}^1 dy \int dx \bar{u} \cdot u = \text{vol}_x \left(\frac{16}{15} + 2\epsilon^2 \int_{-1}^1 (|\partial_y \phi|^2 + \alpha^2 |\phi|^2) dy \right) \quad (\text{A.12})$$

$$+ 2\epsilon^2 \int_{-1}^1 (1 - y^2) \partial_y \delta\phi_0 dy + O(\epsilon)^3. \quad (\text{A.13})$$